Vishvam Patel

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EDUCATION

University of Alabama in Huntsville **Aug 2023 – May 2025**
Master of Science, Computer Science
- **Thesis**: *Neural Acceleration of Graph Partitioning*
- **Relevant Coursework**: Machine Learning, Cloud Computing, Parallel Computing, Numerical Optimization
- **Presentation**: SIAM CSE25 Conference

University of Alabama in Huntsville **Aug 2019 – May 2023**
Bachelor of Science, Computer Science
- Minor: Mathematics
- GPA: 3.67 / 4.0

SKILLS

- **Programming Languages**: Python | C++ | Java | SQL | MATLAB
- **AI/ML Frameworks**: PyTorch | Scikit-Learn | PyTorch Geometric
- **Tools**: Visual Studio | SQL Server | Docker | GitHub | Linux
- **Core Competencies**: Neural Networks | Graph Neural Networks | Optimization Algorithms | Parallel Computing

EXPERIENCE

University of Alabama in Huntsville **May 2023 – May 2025**
Graduate Research Assistant: ML/HPC
- Lead research on neural acceleration techniques for graph partitioning in high-performance and distributed computing environments.
- Developed a neural-based approximation to the Fiedler vector for spectral partitioning, achieving major speedups.
- Built and optimized the Fiduccia–Mattheyses algorithm for efficient bipartitioning with minimized edge cuts.
- Designed experimental ML models using Python, PyTorch, and HPC platforms for efficient graph analysis
- Authored a technical paper submitted to *Journal of Computational Science*

KEY PROJECTS

AdaGrad for Training GNNs: A comparative analysis | Python, PyTorch Geometric, NumPy, Matplotlib
- Conducted a comparative analysis of AdaGrad, Adam, and RMSProp on GNNs over multiple benchmarks.
- Delivered insights on training stability and optimizer selection, supporting model generalization and scalability.

Bank Marketing Random Forest Classifier | Python, Scikit-Learn, Matplotlib, Pandas, NumPy
- Applied EDA, feature engineering, class balancing, and hyperparameter tuning for deposit subscription prediction.
- Achieved 0.95 F1-score with Random Forest using Grid Search and Cross-Validation, demonstrating data-driven model refinement.

Newsgroup Naive Bayes Text Classifier | Python, Scikit-Learn, Matplotlib, NumPy
- Built a full NLP pipeline: tokenization, Gini-based feature selection, and Laplace-smoothed classification.
- Achieved 0.913 F1-score on multi-class text prediction.

Analysis of Optimization Algorithms | MATLAB, Linear Algebra
- Implemented steepest descent, Newton's method, CG, Preconditioned CG, and BFGS for large linear systems.
- Analyzed iteration counts and convergence behavior under different matrix conditions.

LEADERSHIP/HONORS

- Supercomputing Conference (SC'24) - *Student Volunteer*
- UAH Residence Life - *Resident Assistant*
- UAH Office of Admissions - *Student Orientation Leader*
- Pi Kappa Alpha Fraternity - *Vice President of Membership Development*